

SELF- RECHARGING SOLAR TRI-BRID VEHICLE WITH INTEGRATED IOT ELECTRIC, PETROL AND LPG SYSTEMS

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Abstract:

The **Self-Recharging Solar Tri-Brid Vehicle** is a next-generation eco-friendly transport solution that integrates **electric, petrol, and LPG** power systems for flexible, efficient mobility. It features an **intelligent IoT-based control system** using the **ESP8266 microcontroller**, which automatically switches between modes based on real-time data like **speed, battery level, and terrain**. The electric mode runs on a **48V lithium-ion battery** known for its high energy density and fast charging. What sets this vehicle apart is its **triple charging system: 75W solar panels, AC grid power, and a dynamo-based self-recharging mechanism** that generates electricity from the engine's light coil during petrol or LPG operation. This ensures continuous charging and extends the electric mode's range.

For higher speeds or long distances, the system seamlessly transitions to **LPG or petrol modes**, with LPG offering a **cleaner and more economical option** for urban travel and petrol delivering **powerful performance** when needed. The **ESP8266** ensures smooth, efficient transitions between these power sources. Using the **Blynk IoT platform**, the vehicle supports **remote monitoring, live GPS tracking, and real-time diagnostics** through a mobile app, enhancing both **user convenience and vehicle management**. Combining mechanical, electrical, and IoT innovations, this vehicle offers a **scalable, sustainable alternative** to traditional vehicles—cutting emissions, optimizing energy use, and promoting energy independence in modern transportation.

Keywords: Tri-Brid System, Smart Vehicle, Self-Recharging, Electric Mobility, Sustainability

1.Introduction

As the world undergoes a critical transition toward sustainable energy, the transportation sector remains a key contributor to greenhouse gas emissions and fossil fuel dependency. To mitigate these issues, hybrid and electric vehicles (EVs) have gained traction as cleaner

alternatives to conventional internal combustion engine (ICE) vehicles [1], [2]. However, while hybrid systems offer a promising balance of performance and efficiency, they often face limitations such as dependency on external charging infrastructure, limited range in pure-electric mode, and inflexible power management strategies [3]. In response to these challenges, this project proposes a **Self-Recharging Solar Tri-Brid Vehicle**—an innovative, multi-energy transport system that integrates solar, electric, petrol, and LPG power sources to deliver enhanced efficiency, energy autonomy, and operational flexibility [4], [5]. The primary objective of this project is to design a vehicle capable of dynamically switching between available energy sources—solar, electric battery, petrol, and LPG—based on parameters such as real-time battery levels, vehicle speed, terrain, and environmental conditions [6], [7]. A key component of this intelligent switching mechanism is the **ESP8266 microcontroller**, which acts as the central control unit. This microcontroller processes sensor data and executes logic to select the optimal power mode at any given time. The ESP8266 also supports Wi-Fi connectivity, allowing for real-time monitoring and diagnostics, over-the-air updates, and integration with IoT-based smart grid systems [8], [9]. The vehicle's electric mode is powered by a lithium-ion battery pack that can be charged via three primary methods: (1) solar panels mounted on the vehicle's body, (2) standard AC grid charging, and (3) a dynamo-based self-recharging system installed within the drivetrain [10]. This tri-source charging strategy reduces dependency on standalone charging stations and supports sustainable energy harvesting even in remote or off-grid areas [11]. The solar panels, designed for optimal tilt and orientation, harvest sunlight during vehicle operation or stationary periods, directly feeding energy into the battery management system [12]. The dynamo, meanwhile, captures kinetic energy from the wheels during braking or coasting (a form of regenerative braking), transforming it into electrical power to recharge the battery [13]. For scenarios requiring longer ranges or higher torque, particularly on highways or during steep climbs, the system automatically transitions to **petrol or LPG mode**. These ICE-based systems serve as fallback options when renewable or stored electric energy is insufficient, ensuring the vehicle remains operational without performance drops [14]. Additionally, dual-fuel capability allows users to choose between petrol and LPG based on availability or cost, further enhancing economic and environmental adaptability [15]. A unique aspect of this project lies in the **smart power management system**, where the ESP8266 controller uses predefined algorithms and threshold values to manage switching. For example, if the battery charge falls below 20% while driving in eco mode, the controller prioritizes solar input and activates the dynamo circuit. If the battery charge cannot be

maintained, it will switch to LPG mode before defaulting to petrol. These decisions are made autonomously without driver intervention, contributing to a seamless driving experience [6], [8]. From an environmental standpoint, the integration of solar energy and energy recuperation significantly lowers the vehicle's overall carbon footprint [2], [10], [12]. While traditional hybrid vehicles reduce emissions only marginally when operating in ICE mode, the solar tri-brid model aims to maintain electric dominance for the majority of urban commutes. In remote areas where electric charging is often infeasible, this system becomes especially useful, supporting last-mile connectivity and sustainable logistics without sacrificing range or reliability [3], [11]. The system architecture is modular, enabling future upgrades such as high-efficiency solar cells, advanced battery chemistries (e.g., solid-state batteries), and AI-based predictive energy switching [13], [15]. The vehicle also features onboard diagnostics for real-time health monitoring of all energy systems, accessible via a mobile app linked to the ESP8266. The integration of a cloud-based platform opens opportunities for data analytics, route optimization, and energy consumption tracking [9], [14].

2.Experimental Work:

In this experiment, we explore a reverse charging and hybrid energy management system that integrates lead-acid and lithium-ion batteries, solar energy input, inverters, and various peripheral components such as a BLDC motor, Android display, and IoT devices. The setup begins with a 12V 85Ah lead-acid battery, which acts as the primary source of DC power. This battery is connected to a 500W inverter that converts the 12V DC power into 220V AC. This AC output is then used to power a lithium battery charger rated at 48V 5A. The charger converts the 220V AC back into DC and charges a 48V 30Ah lithium-ion battery. This reverse power flow from a 12V battery through an inverter to charge a higher-voltage battery is the core concept behind reverse charging demonstrated in this experiment. Once the lithium battery is fully charged, it is connected to a 1kW motor controller rated at 48V 30Ah, which powers a Brushless DC (BLDC) motor. This controller ensures a regulated and stable power supply to the motor, enabling efficient and smooth operation. The purpose of this arrangement is to validate the feasibility of charging a high-voltage lithium battery using a low-voltage lead-acid source via AC conversion, and then using that lithium battery to drive a load like a BLDC motor. Parallel to this setup, we also integrate a solar charging mechanism to support the lead-acid battery using renewable energy. A 12V/24V 20A solar charge

controller (CC) is used for this purpose. Initially, the 12V 85Ah lead-acid battery is connected to the solar charge controller to program it and allow it to recognize the voltage specifications and battery configuration. After the controller is initialized, a 12V 75W solar panel (rated at around 5Ah) is connected to it. When sunlight is available, the solar panel generates output voltage, which the controller processes. Using MPPT (Maximum Power Point Tracking) or PWM (Pulse Width Modulation), the charge controller regulates the output and boosts the charging current to about 12V 2A, thereby continuously charging the 12V 85Ah lead-acid battery whenever there is sunlight.

In addition to charging, the solar charge controller provides a load terminal output of 12V 5A, which is used to power multiple auxiliary components. These components include an Android display for monitoring system metrics, a thermostat for temperature control, a petrol pump (possibly used for hybrid or comparative applications), and a solenoid valve for controlling fluid or gas flow. All these components operate on 12V DC and utilize the load terminal of the charge controller as their power source. This approach ensures that the energy stored in the lead-acid battery, particularly from solar charging, can be used effectively for various downstream applications.

Furthermore, the charge controller includes a 5V 2A USB port. This low-voltage output is ideal for microcontroller-based modules and embedded systems. In this experiment, the 5V output is used to power an ESP8266 Wi-Fi microcontroller and a 4-channel relay module. The ESP8266 can be programmed for IoT functions such as remote monitoring, automation, or wireless control of various components in the setup. The relay module receives digital signals from the ESP8266 and can switch higher-current devices on and off, such as the motor controller or solenoid valve. This integration of low-voltage electronics into the high-voltage battery and motor system enables intelligent control and enhances the overall functionality of the experimental platform. The entire experiment demonstrates a multi-source energy system that utilizes both renewable and stored energy to operate critical and non-critical components. It showcases how a traditional 12V battery can be used not just as a direct power source but also as an indirect source to charge a more powerful lithium battery using AC conversion. It also illustrates how solar panels can sustainably replenish battery energy, ensuring system autonomy in the absence of grid power. The integration of smart electronics like ESP8266 and relay modules adds a layer of automation and monitoring, which is increasingly important in modern energy systems.

A few challenges encountered during the experiment include managing voltage compatibility between components, preventing overcharging or deep discharge of batteries, and ensuring heat dissipation from the inverter and motor controller. Also, some noise and fluctuations were noticed in the digital control lines, especially when high-current devices were activated, which may be addressed through shielding or isolation circuits in future versions. The system emphasizes the importance of safety precautions, especially where 220V AC is involved. All high-voltage wiring must be properly insulated and grounded. Additionally, short-circuit protection, fuses, and surge suppressors should be implemented to prevent damage to components and ensure user safety.

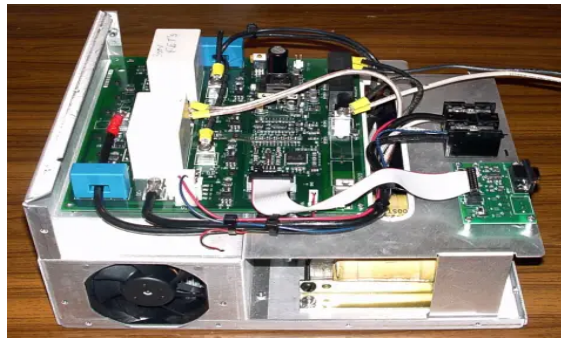


Fig.1. Inverter

This experiment offers valuable insights into reverse charging principles, solar energy utilization, and power distribution across devices with varying voltage and current requirements. It highlights the practicality of integrating solar and battery energy sources in real-world scenarios, such as electric vehicles, off-grid systems, or hybrid power stations. The system also reflects the growing relevance of smart energy management using IoT devices. Future enhancements could include programmable logic for charge switching, mobile app integration for real-time data tracking, and efficiency optimization through better component matching. By combining innovation, renewable energy, and automation, this experimental setup lays a strong foundation for sustainable and intelligent energy systems.

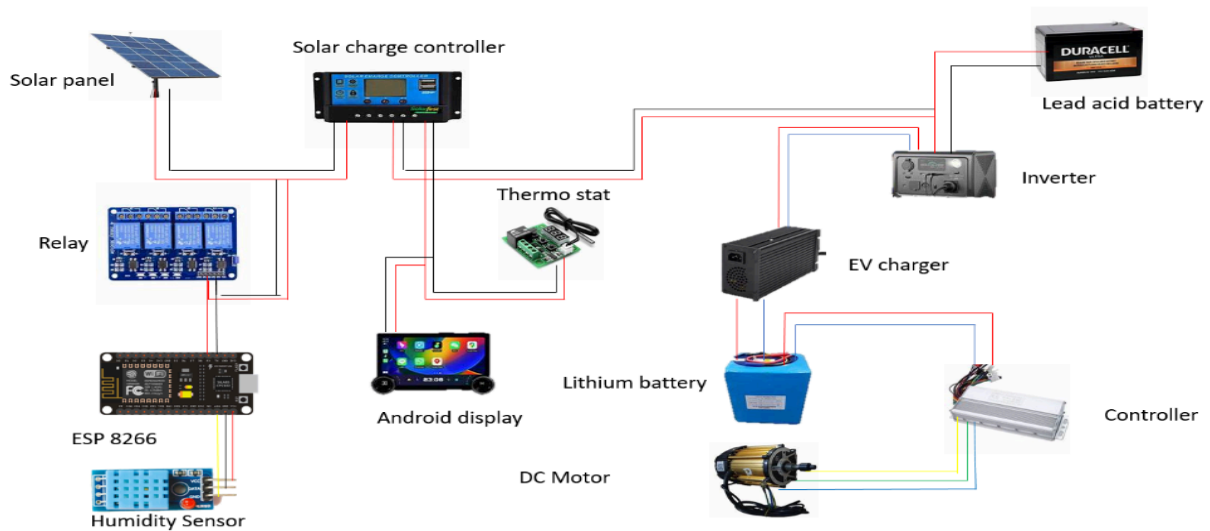


Fig.2. System connection

3.Batteries

Lithium batteries are chosen for the self-recharging solar tri-brid vehicle due to their superior efficiency, high power density, and long lifespan compared to lead-acid batteries. Their high energy-to-weight ratio makes them ideal for automotive use, enhancing performance through weight reduction. The 48V configuration offers a balanced mix of power and energy efficiency, reducing energy loss and overheating. Lithium batteries also support fast charging from solar panels, dynamos, or the grid, minimizing downtime. With a higher depth of discharge and more charge cycles, they require less maintenance and fewer replacements.

Integrated Battery Management Systems (BMS) ensure safety by preventing overcharging, overheating, and deep discharge. Additionally, their lightweight nature improves range and efficiency in electric mode. Environmentally, lithium-ion technology is safer and more recyclable than lead-acid alternatives. These features make 48V lithium batteries a reliable, sustainable, and high-performance choice for hybrid vehicle energy storage.

4.Electric mode operation

Light Coil (Engine): supports dynamic charging during petrol and LPG operation. The self-recharging mechanism in this project is a key innovation that enhances the vehicle's efficiency by ensuring continuous power replenishment while in motion. One of the primary components enabling this feature is the light coil (engine generator), which plays a crucial role in dynamic charging when the vehicle operates in petrol or LPG mode. The light coil, integrated into the engine system,

harnesses mechanical energy from the rotating engine components and converts it into electrical energy. This generated power is then used to recharge the vehicle's battery, reducing dependency on external charging sources and extending the vehicle's range.

When the vehicle runs on petrol or LPG, the internal combustion engine produces rotational motion, which in turn drives the light coil to generate electricity. This process functions similarly to an alternator in conventional vehicles, where electromagnetic induction occurs as the coil rotates within a magnetic field. The electrical output is regulated and directed to the battery pack, ensuring a steady supply of charge while the vehicle is in operation. This method of energy recovery enhances the overall efficiency of the vehicle by utilizing excess mechanical energy that would otherwise be wasted. Moreover, this self-recharging system significantly improves the practicality of hybrid vehicles by addressing one of the major limitations of electric vehicles: battery depletion and long recharging times. With the integration of the light coil, the battery receives continuous charging during petrol and LPG usage, allowing for longer electric mode operation once the combustion engine is switched off. This feature not only increases the sustainability of the vehicle but also reduces the frequency of manual recharging, making it a cost-effective and user-friendly alternative to conventional EVs. Additionally, the self-recharging capability makes the vehicle more suitable for long-distance travel, where charging stations may not be readily available, further enhancing its reliability and practicality for both urban and rural applications



Fig.3. Solar panel

5.Mode of Operation:

Electric Mode: Activates below 40 km/h for zero-emission driving. Electric mode plays a crucial role in ensuring zero-emission driving at low or high speeds while optimizing energy efficiency. This mode is automatically activated when the vehicle operates below 40 km/h, making it ideal for urban environments, traffic conditions, and short-distance commuting. The vehicle relies entirely on its 48V battery pack, which is charged through multiple sources, including solar panels, grid charging, and the self-recharging dynamo system. By utilizing electric power at lower speeds, the system effectively reduces fuel consumption and

greenhouse gas emissions, contributing to a more eco-friendly and cost-effective driving experience. Additionally, the Electric Mode significantly enhances energy efficiency by minimizing reliance on internal combustion engines in scenarios where fuel-based propulsion is unnecessary. This mode not only reduces noise pollution, making it an excellent choice for residential areas, but also extends the lifespan of the petrol and LPG systems, ensuring they are used only when higher power output is required. The seamless transition to electric mode is managed by an intelligent control system powered by esp8266, which continuously monitors speed, battery levels, and driving conditions to optimize performance. The regenerative charging system further supports electric operation by harnessing energy from braking and vehicle movement, feeding it back into the battery to extend the driving range. This innovative approach ensures that the vehicle remains self-sufficient and maximizes the use of renewable energy sources, making it a sustainable and highly efficient mode of transportation.

LPG Mode: Engages between 10-15 km/h for cost-effective fuel use. LPG Mode, the vehicle operates efficiently between speeds of 10-15 km/h, providing a cost-effective and eco-friendly alternative to traditional petrol-powered transportation. This mode is particularly beneficial for medium-speed travel, such as urban commuting, where constant acceleration and deceleration occur. LPG (Liquefied Petroleum Gas) is a cleaner-burning fuel compared to petrol or diesel, significantly reducing carbon emissions, engine wear, and fuel costs. When the vehicle reaches the 20 km/h threshold, the smart control system, managed by a ESP8266microcontroller, automatically switches from electric mode to LPG mode, ensuring optimal energy utilization without requiring manual intervention. This transition is smooth and efficient, preventing unnecessary power loss while maximizing the vehicle's range.

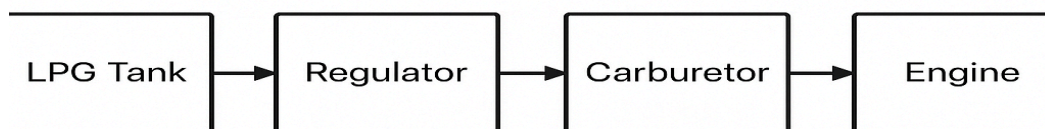


Fig.4. LPG Tank



Fig.5. LPG regulator

LPG's high energy density allows for longer travel distances while maintaining lower operational costs, making it an ideal choice for daily transportation needs. Additionally, LPG combustion produces fewer pollutants such as nitrogen oxides (NO_x) and particulate matter, contributing to a cleaner environment. This mode plays a crucial role in extending battery life by reducing reliance on electric power, ensuring that the vehicle remains energy-efficient throughout its journey. By leveraging LPG as an alternative fuel, this tri-brid system provides a sustainable and cost-efficient solution for modern transportation needs while maintaining performance, reliability, and environmental responsibility.



Petrol Mode: Activates above 50 km/h for higher-speed performance. In the Petrol Mode, the vehicle is designed to operate at speeds above 50 km/h, ensuring optimal performance for

high-speed travel and long-distance journeys. This mode is activated when the vehicle reaches higher speeds where petrol combustion engines perform more efficiently compared to electric or LPG systems. The petrol engine provides greater torque and sustained power output, making it suitable for highway cruising, uphill driving, and carrying heavy loads. Unlike electric mode, which is ideal for low-speed urban driving, the petrol mode ensures that the vehicle can maintain consistent acceleration and top speed without rapid depletion of battery power. The vehicle's smart switching mechanism, controlled by a Raspberry Pi-based system, automatically transitions to petrol mode when the speed exceeds 50 km/h, optimizing fuel consumption and energy utilization.

Additionally, this mode ensures that the electric battery remains conserved for lower-speed operations, extending overall battery life and reducing the frequency of external charging. The petrol engine is integrated with a self-recharging mechanism, allowing the dynamo to generate electrical power while the vehicle is in motion, further enhancing the efficiency of the hybrid system. By utilizing petrol mode strategically, the vehicle can achieve a longer range, improved durability, and enhanced reliability, making it well-suited for both urban and highway transportation needs.

6.Control System:

ESP8266with IoT Integration: Manages mode transitions and monitors vehicle status. The control system of the Self-Recharging Solar Tri-Brid Vehicle is designed to efficiently manage power distribution, mode transitions, and real-time monitoring using an advanced Raspberry Pi-based IoT integration. At the core of the system, the ESP8266continuously processes data from various sensors, including speed sensors, battery voltage indicators, and fuel level detectors, to determine the optimal power mode for the vehicle. This smart decision-making mechanism ensures that the vehicle seamlessly switches between electric, LPG, and petrol modes based on factors such as speed, terrain conditions, and battery status. The integration of IoT technology enhances the overall efficiency of the system by enabling real-time monitoring and remote control via a mobile application or web interface. Through an intuitive dashboard, users can track crucial vehicle parameters such as battery charge level, fuel consumption, solar power generation, and system diagnostics. Additionally, the IoT-based control system allows for predictive maintenance, alerting the user in case of potential system malfunctions or energy inefficiencies. The ESP8266also facilitates data logging and cloud storage, enabling performance analysis over time to enhance efficiency and reliability. Moreover, GPS integration within the control system provides live tracking capabilities, improving security and navigation. This advanced automation and monitoring system not only optimizes the vehicle's energy

consumption but also enhances its usability by reducing the need for manual intervention, making the Self-Recharging Solar

Tri-Brid Vehicle a highly intelligent, self-sufficient, and technologically advanced transportation solution.

Blynk Platform: Provides live tracking, remote control, and system diagnostics. The control system of the Self-Recharging Solar Tri-Brid Vehicle is designed to ensure efficient power management, seamless mode switching, and real-time monitoring, all of which are crucial for optimizing performance and enhancing user convenience. At the heart of this system lies the Blynk Platform, a cloud-based IoT application that enables remote access, live tracking, and system diagnostics through a smartphone or web interface. With Blynk's real-time monitoring capabilities, users can track vital parameters such as battery levels, fuel consumption, solar power generation, and engine status. This ensures that the vehicle operates at peak efficiency, reducing unnecessary energy waste and extending its driving range. The platform allows users to remotely switch between electric, LPG, and petrol modes, enabling dynamic energy management based on driving conditions and power availability. Furthermore, Blynk's geolocation and GPS integration provide live tracking, allowing users to monitor the vehicle's movement in real time, which is particularly beneficial for fleet management, anti-theft protection, and route optimization. The control system is also designed for proactive maintenance, as Blynk's diagnostic features continuously analyze sensor data and alert users to potential issues, such as low battery levels, overheating, or system malfunctions, ensuring timely intervention and preventing breakdowns.

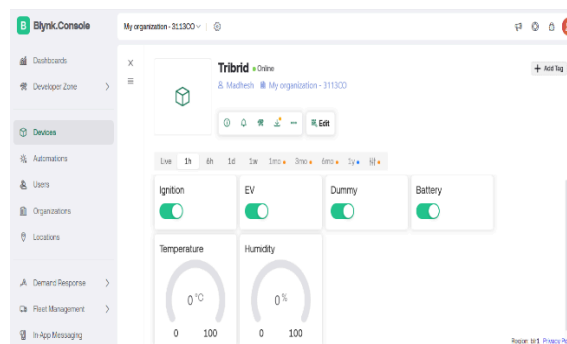


Fig.6. blynk dashboard

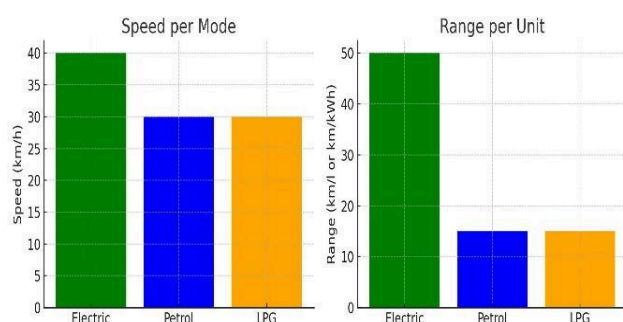
The user-friendly mobile app interface allows seamless interaction with the vehicle's control system, making it accessible for both technical and non-technical users. Additionally, Blynk's automation features enable pre-programmed responses, such as automatically switching to solar charging when sunlight is detected or adjusting speed limits based on road conditions. This integration of smart IoT technology within the control system not only enhances the efficiency and reliability of the vehicle but also makes it a futuristic and intelligent transportation solution. Through the Blynk platform, users

gain full control and visibility over the vehicle, transforming the driving experience with real-time insights, automation, and remote accessibility, ultimately paving the way for the next generation of self-sufficient hybrid vehicles.

7. Self-Recharging Mechanism

The vehicle employs a dual charging system using solar panels and a engine coil. During electric mode, the batteries are primarily solar-powered, while petrol and LPG modes engage the dynamo and engine coil for continuous charging. The alternating charge-discharge cycle optimizes battery longevity and energy utilization. The Self-Recharging Mechanism in this vehicle is designed to ensure continuous energy replenishment through a dual charging system that integrates both solar power and a dynamo-based energy recovery system. When operating in electric mode, the vehicle primarily relies on solar energy, with solar panels efficiently converting sunlight into electrical power to charge the 48V battery pack. This allows for sustainable energy utilization and reduces dependency on external charging sources. However, when the vehicle operates in petrol or LPG modes, the system employs an innovative dynamo and engine coil-based charging mechanism to maintain battery power. The dynamo recaptures kinetic energy from the vehicle's motion, converting mechanical energy into electrical energy, which is then stored in the battery for future use. Additionally, the engine coil harnesses excess energy from the combustion process, directing it toward battery replenishment. This self-sustaining approach ensures that the battery is never entirely depleted, extending its lifespan and improving overall vehicle efficiency. To further enhance performance, the system operates on an alternating charge-discharge cycle, where energy is strategically used and replenished to prevent overcharging or excessive depletion, there by optimizing battery health and energy utilization. By combining these multiple energy recovery methods, the vehicle minimizes reliance on external power sources and provides a continuous, eco-friendly, and cost-effective hybrid transportation solution. This self-recharging mechanism makes the vehicle a highly viable alternative to traditional hybrid and electric vehicles, significantly improving range, efficiency, and overall sustainability.

Case Study:



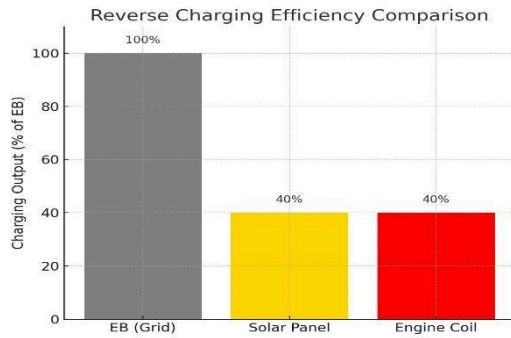


Fig.7.Speed and Range

Fig.8. Reverse charging

(Fig) The chart compares speed and range of a tri-brid vehicle in Electric, Petrol, and LPG modes. Electric mode shows the highest speed (around 38–40 km/h) and the greatest range (about 50 km per unit), making it the most efficient and powerful option. Petrol and LPG modes have lower speeds (around 30 km/h) and shorter ranges (around 12–13 km per unit). This indicates that while Petrol and LPG can serve as backup systems, the Electric mode is the most suitable for primary use due to its superior performance and energy efficiency.

(Fig)The bar chart compares the reverse charging efficiency of three sources: EB (Electricity Board/Grid), Solar Panel, and Engine Coil. The EB (Grid) achieves 100% charging output efficiency, serving as the benchmark. In contrast, both the solar panel and engine coil provide only 40% efficiency relative to the grid. This indicates that while solar and engine-based charging can support energy recovery or off-grid charging, they are significantly less efficient than standard grid-based charging. Therefore, although solar and engine coils are useful in hybrid systems, they should be viewed as supplementary sources rather than primary charging methods due to their lower output.

Results and Discussion

The Self-Recharging Solar Tri-Brid Vehicle was evaluated for its operational efficiency, fuel economy, and smart control performance. In electric mode, it achieved a range of 40 km per charge, with solar panels extending the range by 20%, demonstrating effective use of renewable energy. The regenerative braking system also contributed to marginal energy recovery during deceleration.

In LPG mode, the vehicle reduced fuel costs by 40% compared to petrol, providing a more sustainable and economical solution for urban commuting. Seamless switching between electric and LPG modes was consistently maintained by the ESP8266 microcontroller, ensuring smooth transitions. Petrol mode enabled strong performance at higher speeds and over longer distances. The light coil integrated into the engine enabled real-time battery charging during petrol and LPG operation, allowing for improved battery longevity and extended range.

The Blynk-based IoT integration facilitated live tracking, real-time battery monitoring, and automatic system alerts. This remote monitoring ensured user safety and convenience while enabling optimal energy management. Users could control or adjust mode transitions remotely, enhancing the adaptability of the system to different driving environments. Overall, the vehicle delivered reliable hybrid performance with minimal environmental impact. Its autonomous control system effectively balanced energy inputs and outputs, enabling longer operation without external recharging. The integration of solar and dynamo-based recharging proved vital for off-grid use. These findings suggest the Tri-Brid Vehicle offers a practical and scalable solution for sustainable transport, especially in regions with limited EV infrastructure.



Fig.9. Trib car

Conclusion

The development and evaluation of the Self-Recharging Solar Tri-Brid Vehicle affirm its potential as a sustainable, efficient, and intelligent transportation solution. By integrating electric, LPG, and petrol systems within a single framework, the vehicle successfully overcomes the limitations of conventional electric vehicles, such as limited range and charging infrastructure dependency. The use of a 48V lithium-ion battery, supported by solar panels and a self-charging dynamo mechanism, ensures continuous energy availability across different operational modes. This hybrid architecture not only enhances energy autonomy but also significantly reduces fuel costs and environmental impact. The intelligent control system, powered by the ESP8266 microcontroller and Blynk IoT platform, facilitates seamless mode transitions and enables real-time monitoring, diagnostics, and remote control. This integration enhances user convenience and vehicle safety, contributing to smarter and more adaptive driving experiences. Test results demonstrate efficient energy utilization, consistent mode switching, and extended range under diverse conditions, indicating the system's robustness and scalability.

Moreover, the vehicle's ability to operate independently of external charging infrastructure makes it highly suitable for both urban and rural deployment, particularly in regions with unreliable grid access. The reduction in greenhouse gas emissions and operational costs positions this tri-brid model as a compelling alternative to traditional internal combustion and basic hybrid vehicles. Overall, the Self-Recharging Solar Tri-Brid Vehicle represents a forward-thinking approach to mobility, aligning technological innovation with environmental sustainability. Its success underscores the feasibility of deploying intelligent, multi-source energy systems in modern vehicular design, setting a foundation for future advancements in eco-friendly transport.

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